instead of a BoTorch policy object, we now have a Sobol sequence to generate the set of points that cover our search space, which are then fed into a `MaxPosteriorSampling` object that implements the TS policy. The variable `next_x`, like before, contains the data point that we will query next.

> **NOTE** **Different optimization syntax for Thompson sampling**
>
> **Since we don't have a BoTorch policy object, when visualizing our progress using the `visualize_gp_belief_and_policy()` helper function, we don't specify the `policy` argument anymore. This function therefore will only show the trained Gaussian process at each iteration without the acquisition scores.**

Optimization progress of TS is shown in Figure 5.15, where we can observe that the policy successfully zeros in on the global optimum throughout the procedure, but not without spending a couple of queries exploring the space. This showcases the ability to trade off exploration and exploitation of TS in BayesOpt.

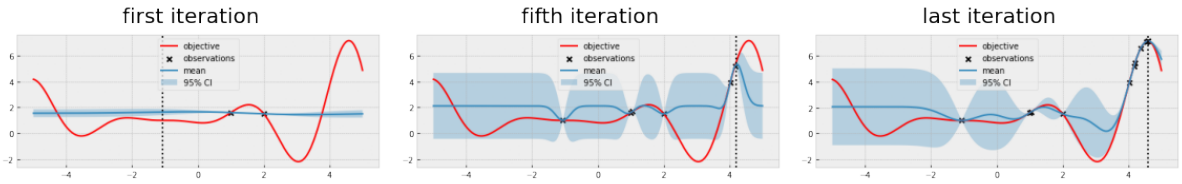


Figure 5.15 Progress made by the Thompson sampling policy. The policy explores the search space for some iterations and then gradually zeros in on the global optimum.

Our discussion on BayesOpt policies that are inspired by policies under the multi-armed bandit setting is thus concluded. We have seen that each of the two policies that we learned, Upper Confidence Bound and Thompson sampling, uses natural heuristics to reason about unknown quantities in an efficient manner and make decisions accordingly. One challenge in BayesOpt, the balance between exploration and exploitation, is also addressed by both policies, allowing the policies to have good optimization performance. In the next and final chapter of the second part of the book, we will learn about yet another commonly used heuristic to decision making in BayesOpt, this time by leveraging information theory.

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